

Connect PHP to MySQL with PDO

Learning Goals

- Distinguish the roles of the PHP server, PDO, and the MySQL database server
- Create the `studentdb` database and `students` table schema with appropriate column types
- Create and configure the dedicated `ase230` MySQL user with least-privilege permissions
- Connect PHP to MySQL using PDO with proper DSN and error handling options
- Execute an `INSERT` statement using the PDO `prepare` and `execute` pattern

Goal: Reach CRUD

CRUD

↑

PHP application logic

↕ PDO

MySQL persistent data

Today, we build that PHP-to-MySQL connection so our application can perform CRUD operations.

IMPORTANT

Who Does What?

CRUD is the four basic operations used to manage stored data: create, read, update, and delete.

- The **PHP server** receives the user's request and runs the application logic.
- **PDO** provides the database abstraction interface connecting PHP to MySQL and carries SQL operations.
- **MySQL** stores, finds, changes, and removes the data.

Create and Select the Database

In the MySQL shell (`mysql>`):

```
CREATE DATABASE IF NOT EXISTS studentdb;  
USE studentdb;
```

- A **database** groups related tables.
- `USE studentdb` selects it for the commands that follow.

```
SHOW DATABASES;
```

Confirm that `studentdb` appears in the result.

The **students** Schema - Database Design

Database design means deciding what information to store and how to organize it before the application saves data.

- A **schema** is the blueprint for a table: its columns, data types, and rules.

The **students** Table Design

Column	Stores	Rule
id	Student identifier	Unique; generated automatically
name , email	Contact information	Text
age , year	Numeric information	Number
major	Field of study	Text

Create the **students** Table

Implement the schema with this SQL command:

```
CREATE TABLE IF NOT EXISTS students (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100),  
    email VARCHAR(150),  
    age INT,  
    major VARCHAR(100),  
    year INT  
);
```

- **Design note on constraints:** In production, required columns like `name` and `email` typically specify `NOT NULL`, so MySQL rejects any `INSERT` that omits them.
 - Here, columns default to nullable for simplicity, so a missing `name` or `email` is silently stored as `NULL` instead of causing an error.

- Why **email** is not **UNIQUE** here: In production, **email** is usually marked **UNIQUE** so MySQL rejects a second row with the same email.
 - Here, **UNIQUE** is intentionally omitted so that refreshing the demo page or re-running the same insert script multiple times keeps adding rows with the same test email, instead of stopping the demo with a duplicate-key error.

Check the Table

```
SHOW TABLES;  
DESCRIBE students;
```

This same student data model will continue into the REST API lesson.

```
id, name, email, age, major, year
```

The database is now ready to support CRUD.

Create a User for the PHP Application

Run these commands in the MySQL shell:

```
CREATE USER IF NOT EXISTS 'ase230'@'localhost'  
  IDENTIFIED BY 'ase230pass';  
ALTER USER 'ase230'@'localhost'  
  IDENTIFIED BY 'ase230pass';  
GRANT SELECT, INSERT, UPDATE, DELETE ON studentdb.*  
  TO 'ase230'@'localhost';
```

IMPORTANT

- Use a separate MySQL account for PHP; do not connect the application as `root`.
- `IF NOT EXISTS` skips creation if the account already exists; `ALTER USER` then resets the password to `ase230pass` regardless, so re-running this block always leaves the account in the same state.

What These User Commands Do

Command	Meaning
<code>CREATE USER</code>	Creates the MySQL account <code>ase230</code> that PHP will use.
<code>IDENTIFIED BY</code> <code>'ase230pass'</code>	Sets the MySQL password for <code>ase230</code> . It is not your macOS, Linux, or browser password.
<code>'ase230'@'localhost'</code>	Allows this account to connect only from the same computer.
<code>GRANT ... ON</code> <code>studentdb.*</code>	Allows the account to perform CRUD operations on every table in <code>studentdb</code> .

Later, PHP sends `ase230` and `ase230pass` to MySQL through PDO. MySQL accepts the connection only when both values match this account.

PDO Connects PHP to MySQL

PDO means *PHP Data Objects*.

PHP code → PDO → MySQL server

In this course, use PDO for every database operation:

```
$pdo = new PDO($dsn, $username, $password);
```

We will use the same `$pdo` connection for Create, Read, Update, and Delete.

Make the PDO Connection

This is the connection section at the beginning of

`index_pdo.php`.

```
<?php
$host = "localhost";
$dbname = "studentdb";
$username = "ase230";
$password = "ase230pass";

$dsn = "mysql:host=$host;dbname=$dbname;charset=utf8mb4";
$pdo = new PDO($dsn, $username, $password, [
    PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
    PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
]);

echo "Connected successfully<br>"
```

Note: Credentials are hardcoded here for local classroom simplicity. Production applications should store credentials in environment variables or configuration files outside the web root.

Run the PHP Server

In your terminal (from the `ase230` course directory):

```
cd module1/code/2_Connect_PHP_with_MySQL  
php -S localhost:8000
```

`php -S` launches PHP's built-in development server, designed for local testing and learning, not for production deployment.

Then open in your browser:

```
http://localhost:8000/index_pdo.php
```

Connection Checkpoint

`index_pdo.php` connects and then inserts one student, so the browser shows:

```
Connected successfully  
Created Student 1 with ID <number>
```

The ID number may be different — it depends on how many students already exist in the table.

If you see this, the complete path works:

Browser → PHP server → PDO → MySQL server

Driver and connection troubleshooting is in document 4 (PDO Troubleshooting, Security Essentials, and MySQLi).

First CRUD Operation: Create

This is the second half of `index_pdo.php` — the same file as the connection above.

```
$sql = "INSERT INTO students (name, email, age, major, year)
      VALUES (:name, :email, :age, :major, :year)";

$stmt = $pdo->prepare($sql);
$stmt->execute([
    ':name' => 'Student 1',
    ':email' => 'student1@example.com',
    ':age' => 20,
    ':major' => 'Computer Science',
    ':year' => 2,
]);
```

IMPORTANT

The PDO Pattern Starts Here

```
$sql = "...";  
$stmt = $pdo->prepare($sql);  
$stmt->execute([ /* values */ ]);
```

You will repeat this pattern for CRUD:

- **INSERT** → Create
- **SELECT** → Read
- **UPDATE** → Update
- **DELETE** → Delete

Next, we will implement all four operations.